

Wheel-hub printed drive sprocket — Rev 010 — 10T on the CUT rim floor (design of record)

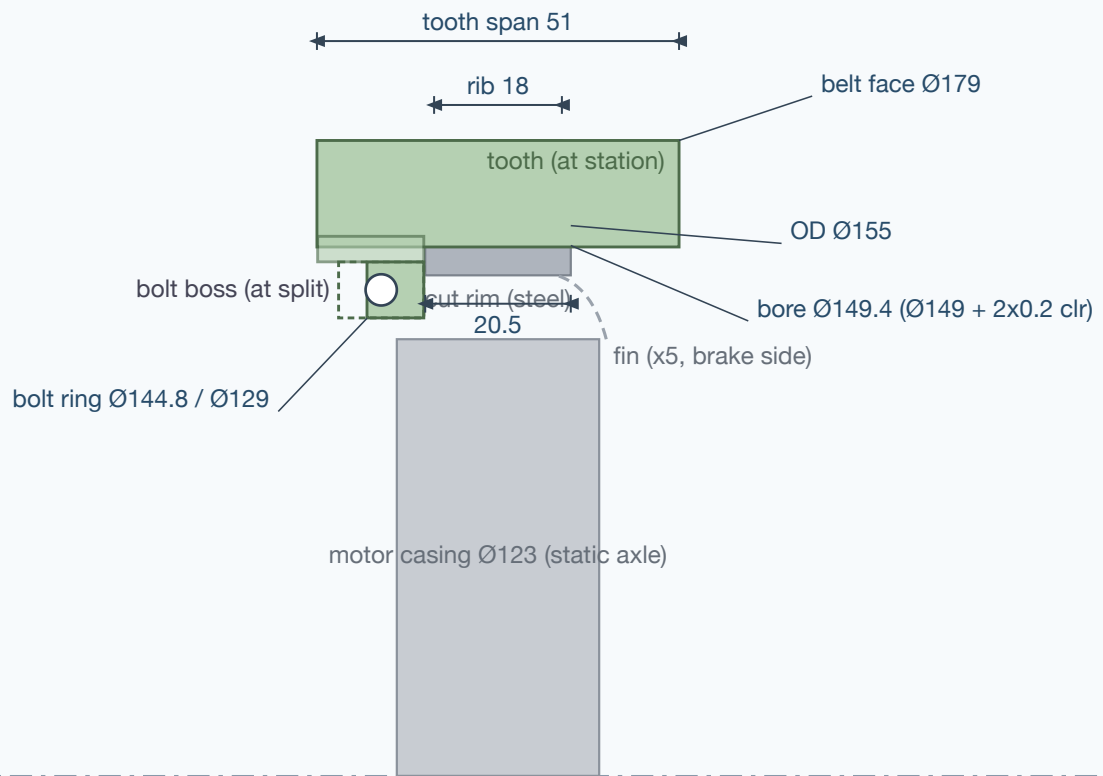
2026-07-28 · companion files: sprocket_print.scad (rev=10) · STLs: stl/rev010_half_A.stl, stl/rev010_half_B.stl, stl/rev010_test_arc.stl · geometry locked to archive/rev010-cut-rim-10T-final

What is new vs the assembly model: the .scad rev models draw the sprocket as rib + teeth only (visual). This blueprint adds the parts that make it a printable, installable component: the drum clamp shell, the two-half split, and an under-floor bolt ring on the **non-fin side** so the halves bolt to each other through the open space between the hub's 5 fins. The bolt ring and bosses are a **new design addition** — verify fin clearance on the real wheel before printing (checklist below).

1 · Key dimensions

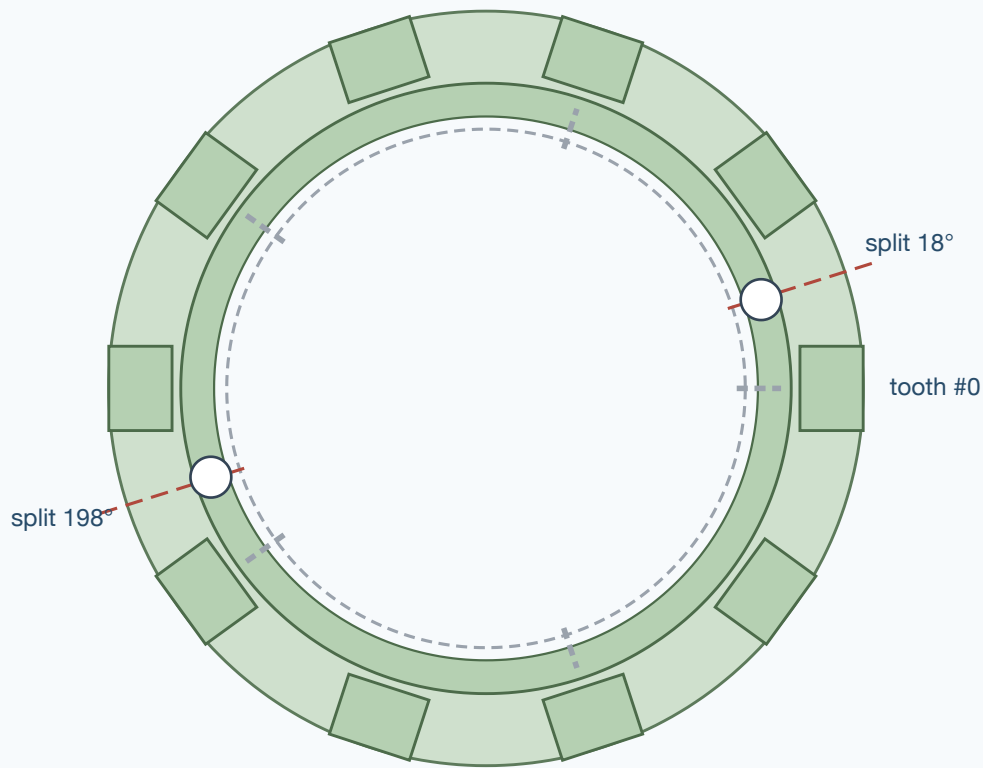
Teeth	10 T-teeth @ 36° — tooth marks every 56.2 mm on the rib top
Belt face (rib top)	Ø179 · cord circle Ø191 (belt: 60 mm pitch, 18 links, T = 12)
Drum seat	bore Ø149.4 = Ø149 floor + 2 x 0.2 fit clearance (tune with test arc) x 20.5 wide
Centre rib	18 wide x 14.99 tall — rides the 22 mm gap between the belt lug pairs
T-tooth	51 across x 20 thick, +5 root widening over the rib zone; outboard blades drop to Ø149 beside the rim
Clamp shell	bore Ø149.4, OD Ø155 (3 mm skin, 2.8 effective after clearance) x 20.5 wide
Bolt ring (NEW)	under-floor ring on the non-fin side: Ø129..Ø144.8 x 8 thick, registers 0.2 off the rim face; tied into every tooth blade
Split	straight split: joint lines at 18° and 198° — both in tooth gaps (36° tooth grid) · 0.25° trimmed per face (~0.3 mm) so the clamp can close
Hardware	2 x M4 x 30 cl.8.8 + nyloc + washers — one per joint line, tangential through the boss pairs (Ø4.4 holes)
Clearances (guarded)	ring OD to belt-lug tip sweep (Ø149): ≥2 mm · ring ID to motor casing (Ø123): 3 mm

2 · Section



half-section through a tooth station · ring side (no fins) on the left · steel = gray ghost, print = green

3 · Face view, split & clocking



face view from the ring side · 10 teeth @ 36° (split lines fall in tooth gaps)
fins (dashed) on a 72° grid — verify both bolt bosses land between fins before printing

4 • Printing

- **Material:** ABS (or ASA). **Orientation:** axis vertical, **bolt-ring face DOWN** — teeth print as continuous layer planes (per the project README).
- **Strength:** ≥ 4 perimeters, $\geq 40\%$ infill (gyroid/cubic), 0.2 layers; no cooling fan for layer adhesion (ABS).
- **Supports:** needed under the shell edge and rib overhangs between tooth stations (the connectors support the part at each tooth).
- **Fit:** `bore_clr = 0.2` per side is the starting value — print the TEST ARC first and adjust for your printer's ABS shrinkage before committing to the halves.

5 • Assembly

1. Degrease the drum seat (the bare ground floor).
2. Dry-fit both halves; the split faces should show a small even gap (~ 0.3 mm/side) — that gap is the clamp stroke.
3. Clock tooth #0 to your reference mark on the hub; confirm both bolt bosses sit between fins.

4. Bolts with washers both sides, nyloc nuts; tighten alternately until the gap closes evenly. Do not crush — snug + a quarter turn (ABS creeps; re-check after the first ride).
5. Spin check: rib runout ≤ 0.5 mm; tooth marks 56.2 mm apart on the rib top.

6 · Pre-print checklist (mandatory)

- Floor ≥ 20 wide and rim ring wall ≥ 3 BEFORE grinding — then grind the flange walls off
- Tape the bare floor: circumference must read 468 mm (= $\varnothing 149$) — send the number before printing
- Print the TEST ARC first — clamp it on the floor; tune bore_clr until snug
- Verify the 5 fins sit at 72.0° nominal and the two bolt bosses (18° / 198° from tooth #0) land between them
- Tooth spacing check on the print: marks every 56.2 mm on the rib top (36°)